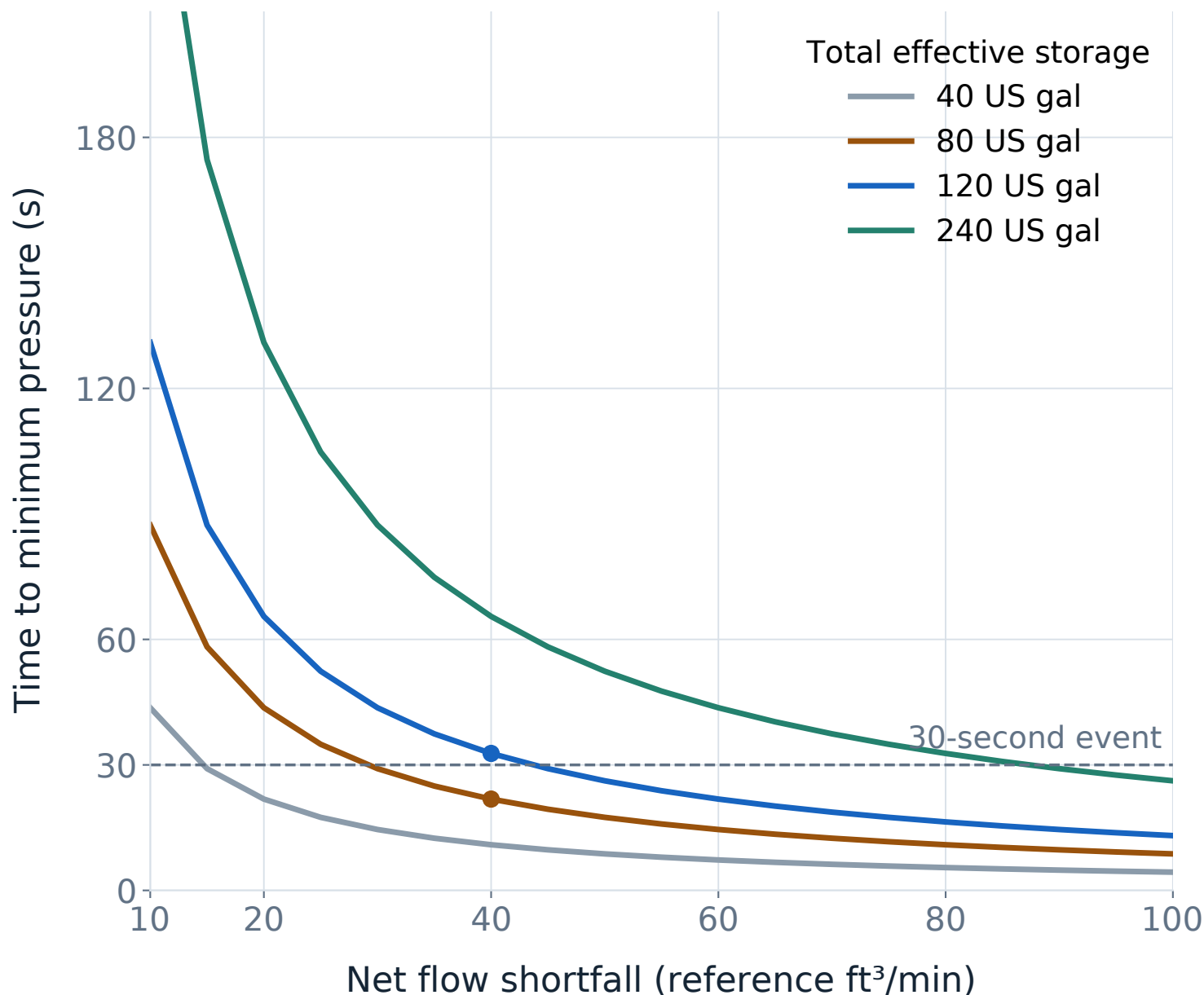


Storage buys time.

How long can storage cover a constant flow shortfall?



At a 40-cfm shortfall:

80 gal provides 21.8 s; 120 gal provides 32.7 s.

Time scales with storage volume and usable pressure band.

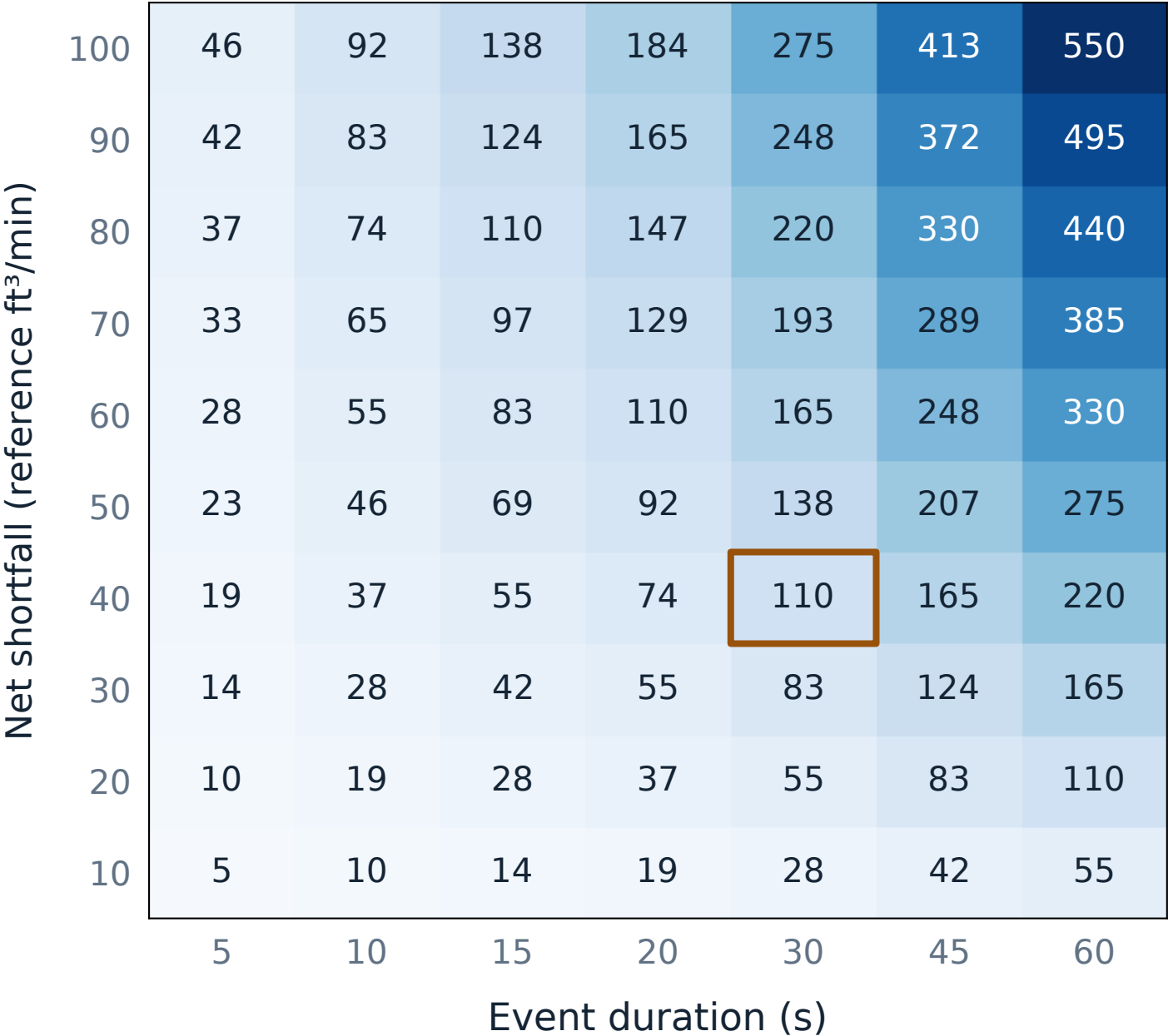
Illustrative inputs · constant-temperature ideal-gas inventory

Flow reference: 1 atm, 20°C · tank: 20°C · pressure band: 20 psi

Method: DOE sourcebook, Fact Sheet 6, p. 43 · AI-assisted analysis

Size the usable buffer.

Required total storage for each shortfall and event duration



Each cell: US gallons, rounded up to a whole gallon.

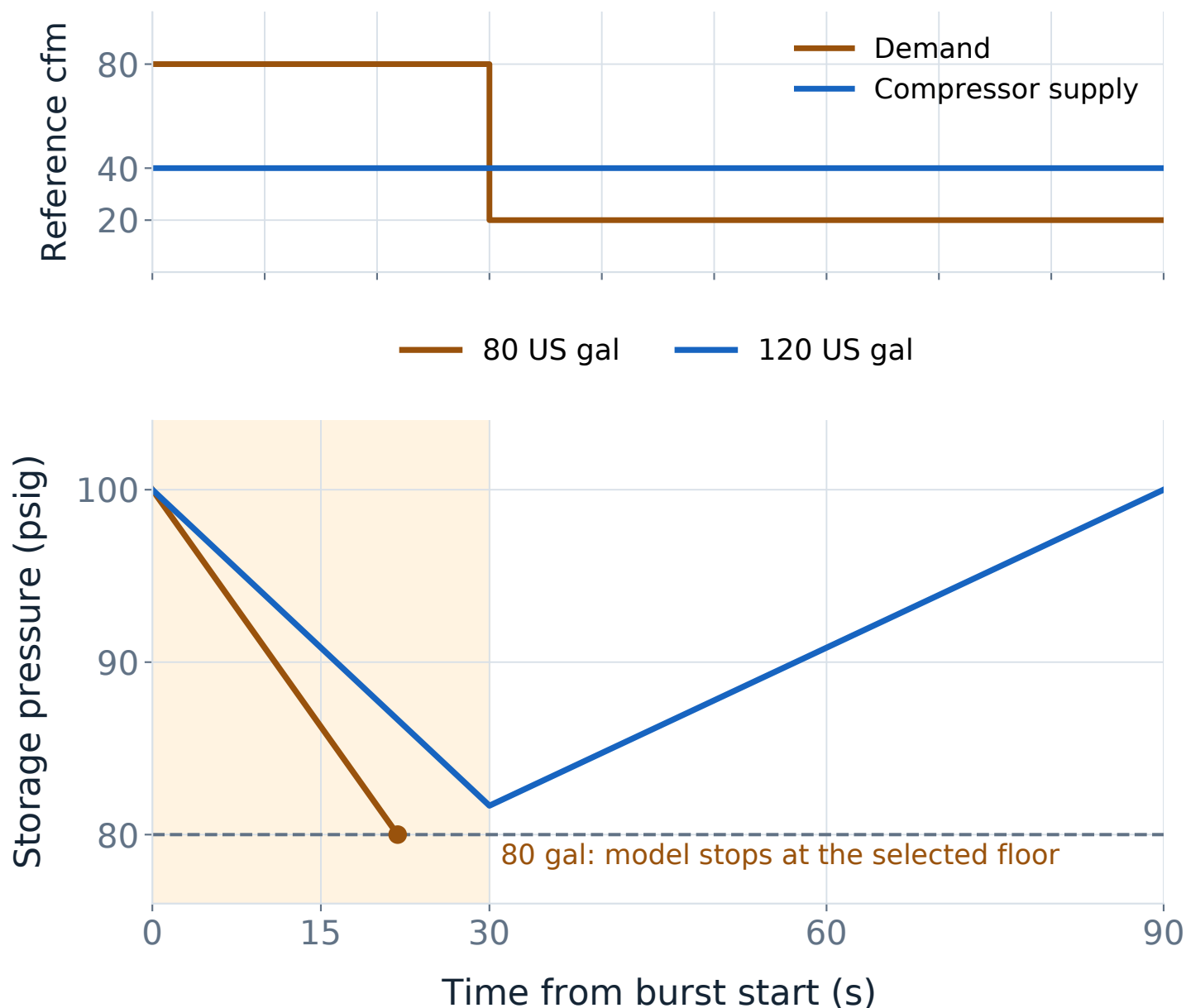
Worked case: 40 cfm × 30 s requires 109.9 US gal.

This is the idealized inventory requirement before design margin.

Illustrative inputs · constant-temperature ideal-gas inventory
Flow reference: 1 atm, 20°C · tank: 20°C · pressure band: 20 psi
Method: DOE sourcebook, Fact Sheet 6, p. 43 · AI-assisted analysis

Check the pressure floor.

A 30-second burst followed by recovery



80 gal reaches the floor after 21.8 s.

120 gal ends the burst at 81.7 psig; recovery takes 60 s.

The burst uses 20 reference ft³; a 20-cfm margin restores it.

Illustrative inputs · constant-temperature ideal-gas inventory

Flow reference: 1 atm, 20°C · tank: 20°C · pressure band: 20 psi

Method: DOE sourcebook, Fact Sheet 6, p. 43 · AI-assisted analysis